

Periodic and Fourier Structure of the Irrational Furstenberg Skew Shift

Route-A structural certificate C169

Abstract

For every irrational α , we solve the affine iterates of $T_\alpha(x, y) = (x + \alpha, y + x)$ on the two-torus. No positive iterate has a fixed point, so the Artin–Mazur zeta is one. On Haar L^2 , the Koopman operator acts explicitly on Fourier modes and admits an involutive reversal. This establishes a sharp contrast between natural operator structure and an empty periodic-orbit ledger.

1 Iterates and periodic points

Induction gives

$$T_\alpha^n(x, y) = \left(x + n\alpha, y + nx + \binom{n}{2}\alpha \right) \pmod{1}.$$

A fixed point would imply $n\alpha = 0 \pmod{1}$, impossible for irrational α . Thus $\#\text{Fix}(T_\alpha^n) = 0$ for every $n \geq 1$ and

$$\zeta_{AM}(z) = \exp\left(\sum_{n \geq 1} \#\text{Fix}(T_\alpha^n) z^n / n\right) = 1.$$

2 Fourier action and reversal

For $e_{m,k} = e^{2\pi i(mx+ky)}$ and $Uf = f \circ T_\alpha$,

$$Ue_{m,k} = e^{2\pi im\alpha} e_{m+k,k}.$$

The $k = 0$ sector is pure point. For $k \neq 0$, residue classes of m modulo $|k|$ give weighted bilateral shifts. Moreover $R(x, y) = (\alpha - x, y)$ obeys $R^2 = I$ and $RT_\alpha R = T_\alpha^{-1}$, so complex conjugation after R is an antiunitary reversal.

3 Boundary

The exact source zeta has no primitive cycles. No target divisor, arithmetic local factor, Hilbert–Pólya operator, or Route-B conclusion is claimed. Scope: NO_BAD_EULER_OR_ROOT_NUMBER.